

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The Modulus and Amplitude of $z = 1 + i\sqrt{3}$ is equal to	CO 1	1
	a. $r = 2, \theta = \frac{\pi}{3}$ b. $r = 2, \theta = 2\pi$ c. $r = 1, \theta = 2\pi$ d. None		
2	The imaginary part of $\sin(x + iy)$ is equal to	CO 1	1
	a. $\sin x \sinh y$ b. $\cos x \sinh y$ c. $-\sin x \sinh y$ d. None		
3	If ω is a complex cube root of unity, then $\frac{1}{\omega} + \frac{1}{\omega^2}$ is equal to	CO 1	1
	a. 1 b. -1 c. 0 d. None		
4	The order and degree of differential equation $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^x$ are	CO 2	1
	a. (1,1) b. (3,3) c. (2,2) d. None		
5	The integrating factor of the Linear differential equation $\frac{dx}{dy} + \left(-\frac{1}{y}\right)x = 2y^2$	CO 2	1
	a. $\frac{1}{y}$ b. $\frac{1}{x}$ c. $\frac{1}{xy}$ d. None		
6	The complementary function of $(D^3 - 6D^2 + 11D - 6)y = 0$ is	CO 3	1
	a. $c_1e^x + c_2e^{2x} + c_3e^{3x}$ b. $c_1e^x + c_2e^{-2x} + c_3e^{3x}$ c. $c_1e^x + (c_2 + c_3x)e^{-2x}$ d. None		
7	The complementary function of $(D^2 - 2D + 1)y = 0$ is	CO 3	1
	a. $c_1e^{-x} + c_2e^{-2x}$ b. $c_1e^x + c_2e^{2x}$ c. $(c_1 + c_2x)e^x$ d. None		
8	The Particular Integral of $(D^2 - 3D + 2)y = e^{3x}$; is	CO 3	1
	a. $\frac{1}{2}e^{3x}$ b. $\frac{1}{4}e^{3x}$ c. $\frac{1}{5}e^{3x}$ d. None		
9	The value of a_0 in the Fourier series expansion of $f(x) = x$ in the interval $(0, 2\pi)$ is	CO 4	1
	a. 2π b. π c. 3π d. None		
10	The value of b_n in the Fourier series expansion of $f(x) = x^2, -\pi \leq x \leq \pi$ is	CO 4	1
	a. 1 b. 0 c. 2 d. None		

11	If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then $\nabla \cdot \vec{r}$ is				CO 5	1
	a. $\vec{3}$	b. $\vec{1}$	c. $\vec{2}$	d. None		
12	The vector field along $\vec{F}$ is said to be irrotational if				CO 5	1
	a. $\nabla \times \vec{F} = 0$	b. $\nabla \times \vec{F} \neq 0$	c. $\nabla \cdot \vec{F} = 0$	d. None		
Q. 2	Solve the following.					12
A)	Find all the values of $\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{3/4}$ and show that their continued product is unity.				CO 1	6
B)	If $\cosh(u + iv) = x + iy$, prove that (i) $\frac{x^2}{\cosh^2 u} + \frac{y^2}{\sinh^2 u} = 1$ (ii) $\frac{x^2}{\cos^2 v} - \frac{y^2}{\sin^2 v} = 1$.				CO 1	6
Q.3	Solve the following.					12
A)	Solve $2y' \cos x + 4y \sin x = \sin 2x$, given $y = 0$ at $x = \frac{\pi}{3}$.				CO 2	6
B)	Solve $ydx - xdy + \log x dx = 0$.				CO 2	6
Q. 4	Solve any TWO of the following.					12
A)	Solve $(D - 2)^2 y = 8(e^{2x} + \sin 2x + x^2)$				CO 3	6
B)	Solve $(D^3 + 1)y = \cos 2x$				CO 3	6
C)	Solve by the method of variation of parameters $y'' - 2y' + 2y = e^x \tan x$.				CO 3	6
Q.5	Solve any TWO of the following.					12
A)	Find the Fourier series of $f(x) = x^2$ in the interval $(0, 2\pi)$				CO 4	6
B)	Find Fourier series for $f(x) = 9 - x^2$ in the range $(-3, 3)$.				CO 4	6
C)	Find half range Fourier cosine series for $f(x) = x$, $0 < x < 2$.				CO 4	6
Q. 6	Solve any TWO of the following.					12
A)	Find $\nabla \cdot \vec{F}$, where $\vec{F} = \left(\frac{x}{r}\right)\hat{i} + \left(\frac{y}{r}\right)\hat{j} + \left(\frac{z}{r}\right)\hat{k}$.				CO 5	6
B)	If $\vec{F} = (x + y + 1)\hat{i} + \hat{j} - (x + y)\hat{k}$, prove that $\vec{F} \cdot \text{Curl } \vec{F} = 0$.				CO 5	6
C)	Using Green's theorem, evaluate $\int_C [(y - \sin x)dx + \cos x dy]$ where C is the plane triangle enclosed by the lines $y = 0$, $x = \frac{\pi}{2}$ and $y = \frac{2}{\pi}x$.				CO 5	6
*** End ***						